

Understanding the LEF File

INTRODUCTION:

This experiment demonstrates the structure and purpose of LEF (Library Exchange Format) files in VLSI physical design. LEF files provide a physical and abstract description of standard cells, including their dimensions, pin positions, symmetry, and routing layers. They serve as a template for place-and-route tools to correctly place and route cells during the physical design process. In this exercise, we will explore a simple LEF file to understand how standard cell characteristics are defined and how this information is utilized in the physical design flow.

APPLICATION:

- Understand the purpose and structure of LEF files.
- Identify cell dimensions, pin names, and layer information.
- Recognize how LEF files guide placement and routing in VLSI design.
- Analyse multiple cells to observe differences in physical attributes.

Step 1: Open Lef File

gvim osu018_stdcells.lef:

Step 2: Search for a Cell

Use / in gvim to locate a standard cell macro

Example: /MACRO AND2_X1

```
VERSION 5.4 ;  
NAMECASESENSITIVE ON ;  
BUSBITCHARS "[]" ;  
DIVIDERCHAR "/" ;  
UNITS  
    DATABASE MICRONS 1000 ;  
END UNITS
```

Step 3: Examine Cell Properties

```
MACRO AND2X1
  CLASS CORE ;
  FOREIGN AND2X1 0.000 0.000 ;
  ORIGIN 0.000 0.000 ;
  SIZE 3.200 BY 10.000 ;
  SYMMETRY X Y ;
  SITE core ;
  PIN A
    DIRECTION INPUT ;
    PORT
      LAYER metall1 ;
      RECT 0.200 3.300 0.600 4.100 ;
    END
  END A
  PIN B
    DIRECTION INPUT ;
    PORT
      LAYER metall1 ;
      RECT 1.300 4.900 1.700 5.700 ;
      RECT 1.000 5.300 1.700 5.700 ;
    END
  END B
  PIN gnd
    DIRECTION INOUT ;
    USE GROUND ;
    SHAPE ABUTMENT ;
    PORT
      LAYER metall1 ;
```

Step 4: Compare Multiple Cells

- Compare AND2_X1, AND2_X2, AND2_X4 for scaling differences in SIZE and pin placement.
- Compare AND3_X1 to observe an additional input pin and its placement.

OUTPUT:

- Identified standard cell sizes, pin positions, and layer information from LEF.
- Observed scaling and layout differences for different drive strengths.

- Understood the placement of power and ground pins using abutment information.

CONCLUSION:

The experiment successfully demonstrated the structure and purpose of LEF (Library Exchange Format) files in VLSI physical design. By opening and analysing the `osu018_stdcells.lef` file using `gvim`, key physical attributes such as cell dimensions, pin positions, symmetry, and routing layer information were identified for the standard cell macro `AND2X1`. Comparing cells `AND2_X1`, `AND2_X2`, and `AND2_X4` revealed that increasing drive strength directly scales the cell size and adjusts pin placement, while `AND3_X1` demonstrated the addition of an extra input pin with its own layer and coordinate definition. The power and ground pins were observed to use `SHAPE ABUTMENT`, confirming that power rails are shared between adjacent cells through row abutment for continuous power delivery. Overall, the experiment provided a clear understanding of how LEF files guide place-and-route tools by providing accurate physical abstractions of standard cells in the physical design flow.